

ANANTHA KRISHNA CHILAPPAGARI

SUMMARY

I am a Machine Learning and Backend Engineer with 5 years of experience building AI systems, cloud-native APIs, and scalable data pipelines. I specialize in Python, AWS, Docker, REST microservices, LLM/RAG workflows, NLP, Gen AI and ML inference services, with hands-on experience improving reliability, cost efficiency, data quality, and operational traceability across distributed systems.

WORK EXPERIENCE

SunPlusData

Atlanta, Georgia

Jan 2025 – Present

Gen AI / Backend Engineer

- Architected containerized GenAI document intelligence services using FastAPI, Docker, YOLO-based layout detection, and LLM workflows to ingest PDFs, extract structured fields, and export validated CSV outputs for SAP integration.
- Built a schema-generation engine that converts user-defined templates into machine-readable extraction contracts, enabling consistent information retrieval across diverse document formats and reducing manual configuration effort.
- Designed SQL metadata and audit schemas for document types, template mappings, file paths, processing states, approval status, and lifecycle history, improving traceability across end-to-end extraction pipelines.
- Implemented JWT authentication and RBAC-driven approval workflows, enforcing fine-grained permissions for viewing, editing, validating, and approving extracted data before downstream submission.
- Developed data quality validation checks for schema compliance, completeness, and accuracy, strengthening pipeline reliability before data moved into business-critical systems.
- Engineered MCP and JSON-RPC tool interfaces that expose REST APIs and stored procedures as LLM-callable tools, enabling natural-language access to operational data with typed inputs, structured outputs, retry logic, and controlled error handling.
- Built ML-based document processing systems that combined Computer Vision and NLP for classification, extraction, and automated understanding of unstructured files.
- Created curated datasets from raw document sources, designing preprocessing, labeling, and validation workflows to support model training and evaluation.
- Trained and fine-tuned ML models for document classification and information extraction, then exposed inference capabilities through Python-based REST APIs.

University of North Texas

Denton, Texas

Jan 2024 – Dec 2024

Instructional Assistant

- Built analytical dashboards for intake trends, department performance, and research output, helping leadership evaluate budget allocation and operational performance.
- Developed a Linear Regression model using budget history, enrollment patterns, and hiring data, reducing unnecessary spending by 15% through data-driven forecasting.
- Mentored students in statistics, machine learning, and applied analytics through labs, project reviews, and hands-on technical guidance.

Constella Intelligence Private Limited

Hyderabad, India

Jan 2021 – Dec 2022

Software Engineer

- Designed event-driven ETL orchestration using AWS Step Functions, Lambda, and CloudWatch, improving reliability and reducing end-to-end processing time by 40%.
- Built containerized ingestion services on AWS ECS with Application Load Balancer autoscaling to support high-volume file intake while reducing idle resource utilization by 20%.
- Developed a Flask REST API for real-time threat assessment with ML-based risk scoring, adding input validation, async request handling, structured logging, and production-ready observability.
- Optimized distributed data processing on AWS EMR with PySpark and automated monitoring, lowering operational costs by 15% and improving failure visibility for production pipelines.
- Contributed NLP feature engineering and model evaluation to an ML-driven cybersecurity pipeline that achieved 93% accuracy in malicious email classification.

PROJECTS

Domain-Specific LLM Fine-Tuning on Finance Dataset — [GitHub Repository](#)

Python, NLP, RoBERTa, FAISS, LLM Evaluation

- Developed a financial QA system by fine-tuning RoBERTa-large on a domain corpus and building a FAISS vector index for semantic retrieval over financial documents.
- Evaluated model performance using ROUGE, BLEU, and BERTScore, comparing fine-tuned models against baseline LLM outputs to quantify retrieval and answer-quality improvements.
- Built a scraping and preprocessing pipeline to collect, clean, and structure financial articles into training-ready datasets for downstream model development.

AI-Driven IT Support & Automation System – AIS Hackathon Runner-Up — [GitHub](#) — [Publication](#)

LangChain, Claude API, MCP, Automation, Reliability Engineering

- Implemented a self-healing monitoring system that detects operational incidents from logs, applies automated fixes, and learns from validated resolutions to improve service reliability.
- Built an AI-driven support assistant using LangChain and Claude API to translate natural-language requests into infrastructure actions through MCP-based tool integrations.
- Integrated SolarWinds automation workflows to perform infrastructure CRUD operations, reducing manual API interactions and accelerating incident response.

Agent-Driven Adaptive Pregnancy Quiz & Flashcard Platform — [Project Live](#)

Python, RAG, LLMs, REST APIs, Docker

- Designed a personalized quiz and flashcard generation system using RAG and LLMs to deliver context-aware, non-repetitive learning content.
- Built Dockerized REST API microservices for adaptive content delivery, user history tracking, spaced repetition, and secure JWT-based profile management.

TECHNICAL SKILLS

Languages:	Python, SQL
AI & ML:	Machine Learning, Deep Learning, NLP, Computer Vision, LLMs, RAG, YOLO, RoBERTa, FAISS, Predictive Modeling, Model Evaluation
Backend & APIs:	FastAPI, Flask, REST API Design, Microservices, JSON-RPC, Model Context Protocol (MCP), Async Processing
Cloud & Distributed Systems:	AWS Step Functions, Lambda, ECS, EMR, CloudWatch, Load Balancing, Autoscaling, Docker, Kubernetes
Data Engineering:	ETL Pipelines, PySpark, SQL Schema Design, Data Warehousing, Metadata Tracking, Data Validation
Security & Reliability:	JWT, OAuth 2.0, RBAC, Rate Limiting, Retry Logic, Structured Error Handling, Logging, Monitoring
DevOps & Quality:	Git, GitHub Actions, CI/CD, Pytest, Code Reviews, Source Control, Testing Workflows

EDUCATION

University of North Texas Master of Science in Data Science	Aug 2023 – May 2025
CGPA: 3.99/4.0	
Jawaharlal Nehru Technological University B.Tech in Computer Science	2018 – 2022
CGPA: 3.5/4.0	